

A/B Test Analysis for an E-Commerce Website

Comparing Two Pricing Strategies Using Hypothesis Testing

Mini Project Report

Abstract

*This project evaluates whether a new pricing/landing page (the "treatment") converts website visitors better than the existing page (the "control"), using a real e-commerce A/B test dataset of 294,478 user sessions. After cleaning the data (removing mismatched group/page records and duplicate users), 290,584 sessions across three countries (United States, United Kingdom, Canada) were analyzed. Three complementary statistical tests were applied: an independent-samples t-test, a chi-square test of independence, and one-way ANOVA. All three tests agree: there is **no statistically significant difference***

*in conversion rate between the two pricing strategies ($p \approx 0.190$), and this holds even when the data is broken down by country. Based on this evidence, the recommendation is to **retain the existing pricing page** rather than incur the cost and risk of switching to a new page that shows no measurable benefit.*

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1. Introduction

E-commerce companies routinely test changes to their websites before rolling them out to every customer. This project analyzes a controlled experiment in which visitors to an e-commerce site were randomly assigned to one of two groups:

- **Control group** shown the existing ("old") pricing page.
- **Treatment group** shown a new pricing page.

For every visit, the dataset records whether the user converted (made a purchase, coded as 1) or not (coded as 0). The business questions this report answers is: Does the new pricing strategy produce a higher conversion rate than the old one, with statistical confidence—or is any observed difference simply due to chance?

To answer this rigorously, three classical hypothesis-testing techniques are applied and cross-checked against one another:

1. An independent two-sample t-test comparing mean conversion between groups.
2. A chi-square test of independence on the group conversion contingency table.
3. A one-way ANOVA testing whether conversion differs across countries, and across the six country/strategy combinations.

2. Data Description

Two files were provided:

- **ab_data.csv**: 294,478 rows, with columns `user_id`, `timestamp`, `group`, `landing page`, and `converted`.
- **countries.csv**: 290,584 rows mapping user id to a country (US, UK, or CA).

2.1 Data Cleaning

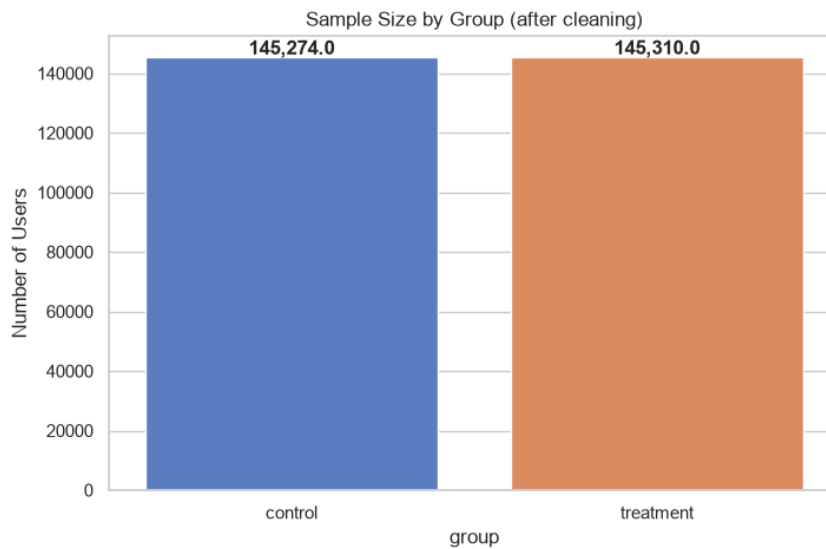
Two data-quality issues were identified and corrected before analysis:

1. **Group/page mismatches**. 3,893 rows violated the rule where control users see old page and treatment users see new page and were removed.
2. **Duplicate users**. 3,894 `user_id` values appeared more than once. Only the first recorded visit for each user was retained.

After cleaning and merging with the country data, 290,584 sessions remained, split almost evenly between the two groups (Figure 1).

Group	Users (n)	Conversions
Control (old page)	145,274	17,489
Treatment (new page)	145,310	17,264
Total	290,584	34,753

Table 1: Sample sizes after cleaning



3. Methodology

All tests use a significance threshold of $\alpha = 0.05$. For each test, the null hypothesis (H_0) represents "no effect / no relationship," and the alternative hypothesis (H_1) represents "there is an effect." A p-value below 0.05 leads us to reject H_0 in favor of H_1 .

- **3.1 Test 1:** Independent t-test: Welch's independent two-sample t-test was used to compare the mean conversion rate of the treatment group against the control group.
- **3.2 Test 2:** Chi-square test of independence: A chi-square test on the 2x2 contingency table (group vs. converted) provides a check on whether conversion depends on which page a visitor saw.
- **3.3 Test 3:** One-way ANOVA: Compared mean conversion rates across the three countries (US, UK, CA), and separately across the six country-strategy combinations.

4. Results

4.1 Descriptive Overview

The control group converted at 12.04% and the treatment group at 11.88%—a raw difference of just -0.158 percentage points in favor of the old page.

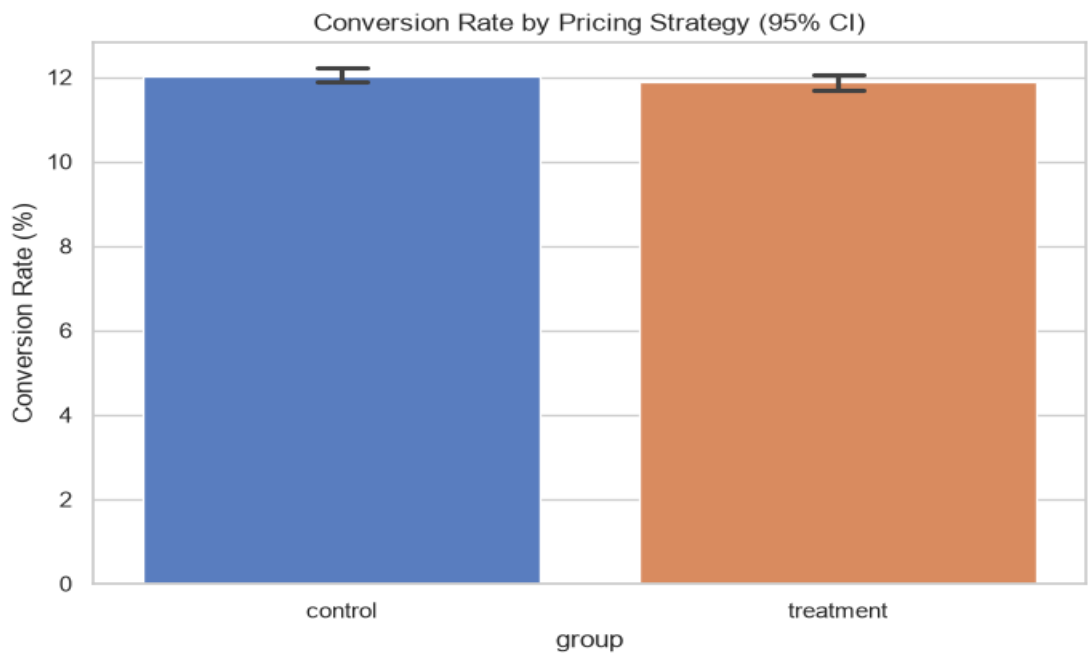


Figure 2: Conversion rate by pricing strategy. The difference is statistically negligible.

4.2 T-Test Results

Metric	Value
Control mean conversion	12.04%
Treatment mean conversion	11.88%
Difference (treatment - control)	-0.158 pp
t-statistic	-1.311
p-value	0.190

4.3 Chi-Square Test Results

Group	Not Converted (0)	Converted (1)
Control	127,785	17,489

Treatment	128,046	17,264
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Table 3: Contingency table: strategy vs. conversion outcome

Metric	Value
Chi-square statistic (χ^2)	1.704
Degrees of freedom	1
p-value	0.192

Table 4: Chi-square test of independence

The chi-square test agrees with the t-test: $p = 0.192 > 0.05$, so we fail to reject H_0 . The observed counts are very close to what would be expected if conversion were completely independent of which page a visitor saw.

4.4 ANOVA Results

Comparison	F-statistic	p-value
Conversion across 3 countries	1.605	0.201
Conversion across 6 country-strategy groups	1.466	0.197

Table 5: One-way ANOVA results

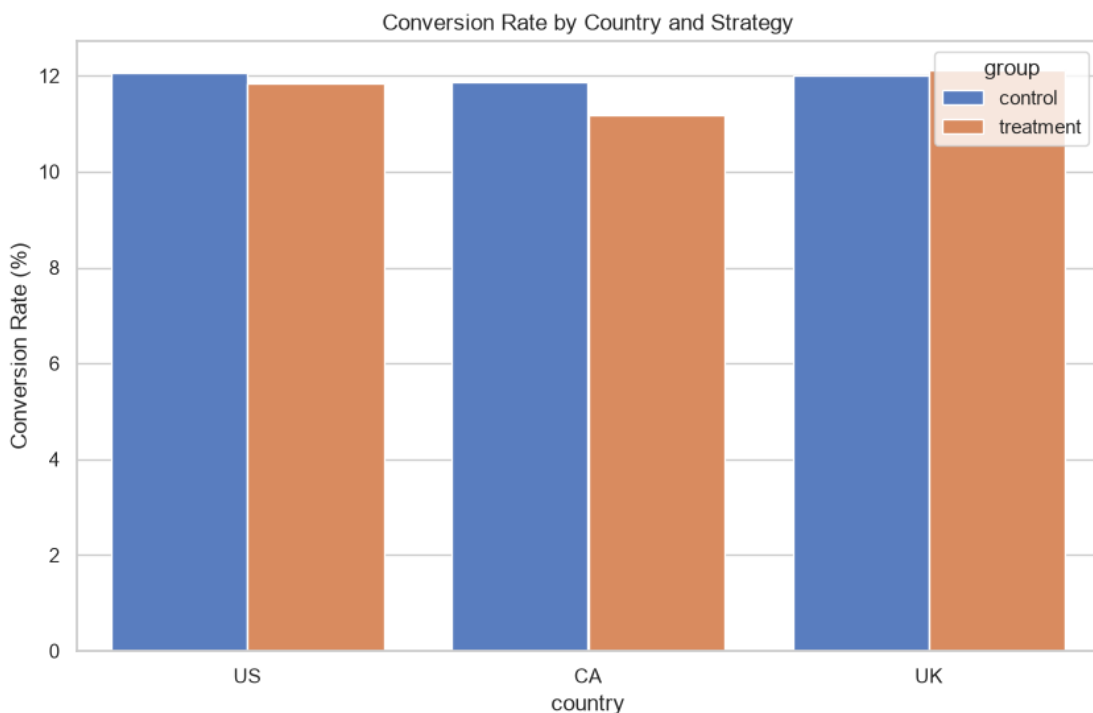


Figure 3: Conversion rate by country and strategy. No country shows a consistent advantage for either page.

Both ANOVA results give $p > 0.05$, so we fail to reject H_0 in each case. Conversion rate does not differ significantly across countries, and no specific country/strategy combination stands out as a hidden winner.

5. Discussion

All three tests—despite examining the data from different statistical angles—reach the same conclusion, which is a good sign of robustness rather than a coincidence:

- The t-test shows **no significant difference** in mean conversion between the two pricing pages.
- The chi-square test shows conversion is **statistically independent** of which page a visitor saw.
- The ANOVA shows the picture **doesn't change** when the population is split by country—there is no hidden sub-population where the new page clearly wins.

6. Conclusion and Recommendation

Based on rigorous statistical testing across three independent methods, there is NO EVIDENCE that the new pricing page improves conversion over the existing page, either overall or within any individual country.

Recommendation: The company should RETAIN THE CURRENT PAGE. All analysis was performed in Python using pandas for data handling and scipy.stats for hypothesis testing.